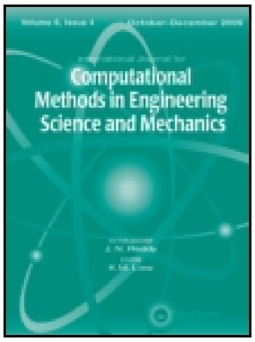

International Journal for Computational Methods in Engineering Science and Mechanics

ISSN: 1550-2287 (Print) 1550-2295 (Online) Journal homepage: www.tandfonline.com/journals/ucme20

An Optimal Design Methodology of Tapered Roller Bearings Using Genetic Algorithms

Rajiv Tiwari, Kumar K. Sunil & R. S. Reddy

To cite this article: Rajiv Tiwari, Kumar K. Sunil & R. S. Reddy (2012) An Optimal Design Methodology of Tapered Roller Bearings Using Genetic Algorithms, International Journal for Computational Methods in Engineering Science and Mechanics, 13:2, 108-127, DOI: 10.1080/15502287.2011.654375

To link to this article: https://doi.org/10.1080/15502287.2011.654375

Published online: 13 Mar 2012.

International Journal for Computational Methods in Engineering Science and Mechanics, 13:108–127, 2012

ISSN: 1550-2287 print / 1550-2295 online
DOI: 10.1080/15502287.2011.654375

An Optimal Design Methodology of Tapered Roller Bearings Using Genetic Algorithms

Rajiv Tiwari, Kumar K. Sunil, and R. S. Reddy
Department of Mechanical Engineering, Indian Institute of Technology Guwahati, Guwahati, India

In the design of tapered roller bearings, long life is the one of the most important criterion. The design of bearings has to satisfy constraints of geometry and strength, while operating at its rated speed. An optimal design methodology is needed to achieve this objective (i.e., the maximization of the fatigue life). The fatigue life is directly proportional to the dynamic capacity; hence, for the present case, the latter has been chosen as the objective function. It has been optimized by using a constrained nonlinear formulation with real-coded genetic algorithms. Design variables for the bearing include four geometrical parameters: the bearing pitch diameter, the diameter of the roller, the effective length of the roller, and the number of rollers. These directly affect the dynamic capacity of tapered roller bearings. In addition to these, another five design constraint constants are included, which indirectly affect the basic dynamic capacity of tapered roller bearings. The five design constraint constants have been given bounds based on the parametric studies through initial optimization runs. There is good agreement between the optimized and standard bearings in respect to the basic dynamic capacity. A convergence study has been carried out to ensure the global optimum point in the design. A sensitivity analysis of various design parameters, using the Monte Carlo simulation method, has been performed to see changes in the dynamic capacity. Illustrations show that none of the geometric design parameters have adverse affect on the dynamic capacity.

Keywords Tapered roller bearings, Optimum design, Genetic algorithms, Parametric sensitivity analysis, Monte Carlo simulations

1. INTRODUCTION

Tapered roller bearings typically have higher radial load capacity than ball bearings; however, the former has a low axial capacity and higher friction under axial loads. If the inner and outer races are misaligned, the bearing capacity often drops quickly as compared to either a ball bearing or a spherical roller bearing. Tapered roller bearings use conical rollers (i.e., frustum of a cone) that run on conical races. Tapered roller bearings have the ability to carry a combination of the large radial and thrust loads, or to carry the thrust load at moderate speeds, and thus have some of the same advantages as those of ball bearings. Because of this, tapered roller bearings are exploited in many automobile applications, such as in the car front and rear wheels, machine tool spindles, locomotive wheel axles, rolling mills, construction machines, lifting machines, printing machines, and various decelerating devices. However, due to relatively large sliding friction generated at the flange, the tapered roller bearing is not suitable for high-speed operation unless a special arrangement for the cooling and/or lubrication is made. By equipping the bearing with specially contoured flanges, a special cage or retainer, and lubrication holes, a tapered roller bearing can be used with the high reliability under the high-load, high-speed conditions for automobile and aircraft applications. In such applications, the demand of high reliability has given a boost to the advancement in design technology of taper roller bearings. This has motivated design engineers to come up with design technologies that will optimize the fatigue life of the bearing. Before making an attempt at detailed analysis of optimum design of taper roller bearings, understanding the basic geometry of taper roller bearings is necessary.

A plethora of research work is available on the design and analysis of a variety of machine elements; however, very few studies are available on rolling element bearings, especially on tapered roller bearings. Moyer and Mckelvey [28] studied the rating formula used by the Timken roller bearing company to determine basic ratings for tapered roller bearings. Karna [25] presented both analytical and experimental investigations to determine the nature of the coefficient of friction at the rib-roller end of a tapered roller bearing. Andreason [1] described a method to calculate bearing loads in a two-row tapered roller bearing arrangement; he considered the misalignment, the initial preload, and deflections of the shaft and the housing. Liu [26] presented an analytical study of the load distribution in tapered roller bearings, operating at high speeds and combined loadings, without considering friction forces. Changsen [5] described an optimization procedure for the design of tapered roller bearings by taking the dynamic load capacity as the objective function with

Address correspondence to Rajiv Tiwari, Department of Mechanical Engineering, Indian Institute of Technology Guwahati, Guwahati 781039, India. E-mail: rtiwari@iitg.ernet.in

associated geometric constraints. Constraints contain constants, which were suggested to be chosen arbitrarily. This makes the understanding of the effect of the individual constraint difficult and unrealistic. A few of the selected design parameters were not directly affected by the dynamic load capacity. Moreover, no illustrations were provided and only the procedure was described. Harris [18] gave a comprehensive analysis of rolling element bearings. Apart from the simplest elements of rolling bearings, such as bearing types, geometry, tolerances, etc., it also deals with the complex analysis of the static and dynamic load distributions of a variety of rolling bearings.

The design of rolling element bearings is a nonlinear optimization problem with associated constraints. There are many numerical deterministic techniques available for solving the nonlinear constrained optimization problems. However, the main difficulty of conventional deterministic methods is slow convergence along with local optimal (or sub-optimal) solution. These problems are more serious when the number of parameters to be optimized is fairly large. Genetic algorithm (GA) is a non-conventional method, which is suitable for nonlinear optimization problems, and in most cases they can find the global optimal solution with a high probability. Rao [30] described different optimization techniques applied to engineering problems. Deb [10] discussed the working of different conventional optimization search techniques in the form of algorithms. Conventional optimization techniques are suitable for specific problems only. Non-traditional search techniques were developed to solve the problems that are difficult to attempt with conventional methods. Holland [20] studied adaptation in the natural and artificial systems and invented genetic algorithms (GAs) to solve complex engineering problems where conventional methods were not suitable. Goldberg [16] described the working principle of genetic algorithms and concluded that algorithms are mathematically simple, yet powerful in their search for the improvement of objective function after each generation. In standard GAs, the binary coding of variables are used, which have inherent disadvantages such that the computational complexity increases with the increase in the desired accuracy. To overcome this difficulty, GAs have also been developed to work directly with continuous variables, known as the real coded genetic algorithm (RCGA) [10, 15]. Chaturvedi et al. [8] applied RCGAs for structural optimization problems. Chakraborthy et al. [6] and Chaturbhuj et al. [7] optimized the design of deep-groove bearings and tapered-roller bearings, respectively, using genetic algorithms, and they showed that the life of the bearing improved moderately with the life provided in standard bearing catalogues. However, some of the optimization constraints they formulated were unrealistic. Recently, Rao and Tiwari [31] and Sunil Kumar et al. [33, 34] improved the design optimization of, respectively, deep-groove and crowned cylindrical roller bearings using genetic algorithms, and the major improvement was in formulating realistic constraints.

In the present work, a constrained nonlinear optimization problem for the design of tapered roller bearings has been formulated and a procedure for solving the optimization problem has been described. The main effort has been put into formulating realistic constraints based on geometric, bearing standards, and strength considerations. Geometric constraints are formulated from standard boundary dimensions in bearing standards. Strength constraints have been formulated considering the contact stresses and the subsurface shear stresses at contacts of the roller with the cup, cone, and flange. The load distribution is obtained by Stribeck's (conservative) formula [18], which is used to obtain maximum loads on the flange and the rings. All the constraints formulated are satisfied before the acceptability of the design variables for calculation of the fatigue life during the convergence. The input for the design problem has been selected from the bearing standards and catalogues. By using real-coded genetic algorithms, the design optimization problem has been optimized and the corresponding dynamic load capacity shows improvement in the fatigue life as compared with that given in standard catalogues. A sensitivity analysis has been performed to observe the effect of manufacturing tolerances on design variables.

2. MACRO-GEOMETRY OF TAPERED ROLLER BEARINGS

A tapered roller bearing, as shown in Figure 1, comprises an inner ring (or cone), an outer ring (or cup), and a compliment of rollers in the shape of truncated cones. Contact lines between rollers and raceways intersect at a common point on the axis of the bearing to ensure that the relative motion between rollers and raceways approaches true rolling. A cage retains the set of rollers on the cone as a single unit. The cup can be withdrawn from the bearing assembly without any effort. Due to the inclined position of the raceways, a single row tapered roller bearing generally cannot accept a pure radial load. Hence, it must also be subjected to the axial load or it must be preloaded or supported axially.

In tapered roller bearings, the cone has two lips, namely the smaller and larger lips, and they have different functions. In conjunction with the cage, the smaller lip retains rollers on the cone raceway, whereas the larger lip carries the axial force com-

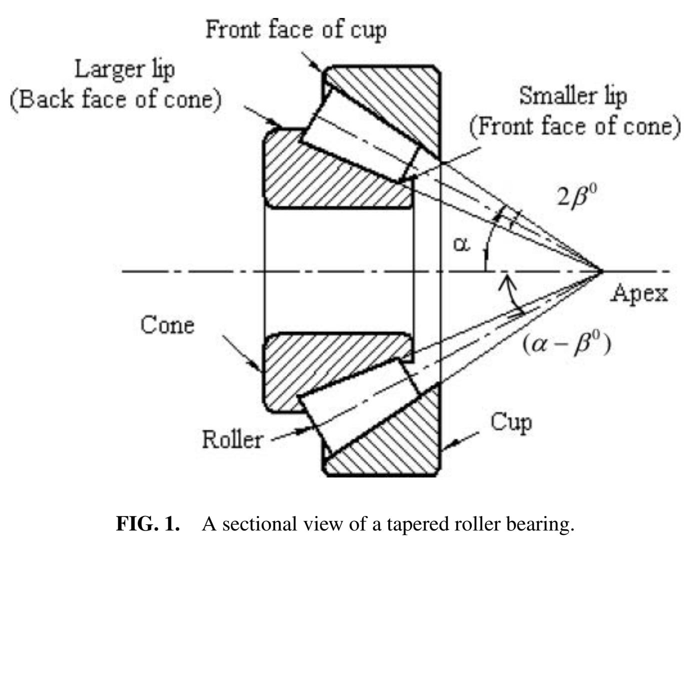

FIG. 1. A sectional view of a tapered roller bearing.

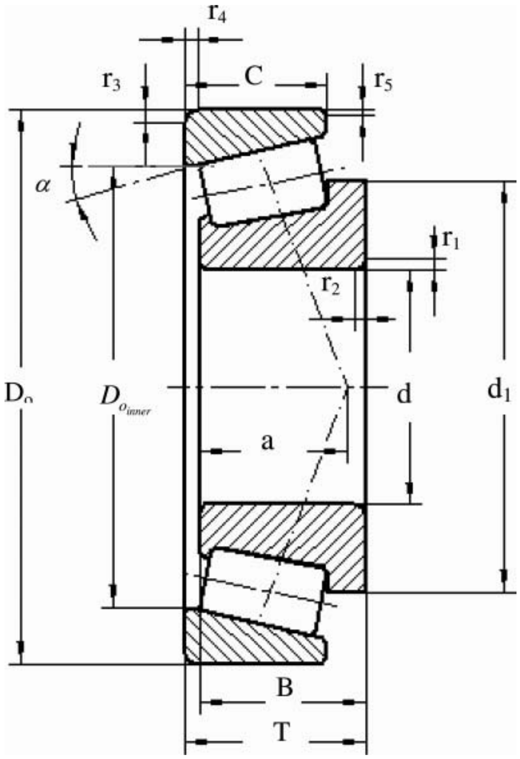

FIG. 2. Nomenclature of single row tapered roller bearing with the boundary dimensions of BIS.

ponent resulting from the tapered shape of rollers. As tapered rollers run on the roller path, they slide along the larger lip and a large amount of the power loss occurs. An essential requirement for the design of a bearing is the three-dimensional mounting space available for the bearing, which is decided by the standard boundary dimensions. These dimensions are the input for the design problem. Figure 2 shows the nomenclature of a single row tapered roller bearing with the specification of boundary dimensions as per bearing standard IS 7461 [3] and listed in the Nomenclature. The contact angle, α, is unique for a set of boundary dimensions of a metric series tapered roller bearing and has been selected from the bearing standard [3] along with boundary dimensions To avoid sharp edges, the corners of faces of the cup and the cone are chamfered as shown in Figure 2. The minimum value of these chamfer heights and widths are provided in the bearing standards and catalogues along with basic boundary dimensions. To select boundary dimensions for the present problem, the bearing standard [3] and the SKF bearing general catalogues [32] have been consulted.

3. DESIGN OPTIMIZATION OF TAPERED ROLLER BEARINGS

Design optimization of the tapered roller bearing is formed as a nonlinear constrained optimization problem. It can be solved using stochastic optimization methods. The optimization technique used for the present problem is a single objective, real coded GAs; it requires a design objective function to optimize. On the basis of operating requirements, different objective functions for the rolling element bearing may be proposed. In normal operating conditions, the rolling bearings have different modes of failure, such as contact fatigues, corrosions, over-heating, loss of accuracy, increase in friction moments, severe vibrations, and noise. Some of the failures can be avoided by adapting proper design and maintenance [5]. However, the contact fatigue is the most critical mode of failure in rolling bearings, which cannot be avoided completely. A basic requirement for conventional rolling bearings is the longest fatigue life. Designing of roller bearings in achieving the longest fatigue is a big challenge to designers. Therefore, bearings are often designed for maximum fatigue life as the objective function. The fatigue life of a bearing can be expressed as

$$L_{10} = \left(\frac{C_d}{P}\right)^n 10^6 \tag{1}$$

where L_{10} is the life of a bearing in millions of revolutions with 90% reliability, C_d is the dynamic load capacity, P is the equivalent radial load, and n is the load life exponent that is taken as 10/3 for the line contact [18]. The fatigue life equation (1) is a function of dynamic load capacity C_d and the equivalent radial load P. For the maximum fatigue life of a bearing, the dynamic capacity C_d should be maximized for a given size of bearing outline and for a particular bearing load. In a subsequent subsection, the objective function and constraints will be formulated for the optimum design of tapered roller bearings.

3.1 Objective Function

The objective function is applied to evaluate the “goodness” of design optimization of a tapered roller bearing. The objective function of the optimization problem is also known as a fitness function for GAs; this objective function prescribes the optimality of a solution in the GA technique. The basic dynamic load capacity is the most critical factor in deciding the fatigue life of a bearing. From equation (1) it can be known that the fatigue life of a bearing is directly proportional to the dynamic capacity. In the present optimization problem, maximization of the fatigue life of the bearing is the main objective function. Hence, it is given as

$$\max\,[f(X)] = \max\,[C_d] \tag{2}$$

where $f(X)$ represents the objective function and X is the design variable vector. The expression for the basic dynamic load capacity for roller bearings is given as [18]

$$C_d = b_m f_c\,(i l_e \cos\alpha)^{\frac{7}{9}}\, Z^{\frac{3}{4}}\, D_r^{\frac{29}{27}} \tag{3}$$

with

$$f_c = 207.9\lambda\nu\gamma^{\frac{2}{9}}\frac{(1-\gamma)^{29/27}}{(1+\gamma)^{1/4}} \times \left[1+\left\{1.04\left(\frac{1-\gamma}{1+\gamma}\right)^{\frac{143}{108}}\right\}^{9/2}\right]^{-2/9} \tag{4}$$

$$\gamma = \frac{D_{r_{mean}}}{D_m}\cos\alpha \quad \text{and} \quad D_{r_{mean}} = \frac{1}{2}(D_{r_{L.L}} + D_{r_{U.L}}) \quad (5)$$

From equation (3) it is clear that the basic dynamic capacity depends upon the (7/9)th to the power of effective length of roller (l_e), (3/4)th to the power of number of rollers (Z), and (29/27)th to the power of mean roller diameter (D_r). Hence, during the optimization, we expect that the maximum possible roller diameter would give us better dynamic capacity. Moreover, with the bigger roller diameter, a reduced number of rollers would be accommodated in a given space. The dynamic capacity is derived on the basis of the maximum contact stresses that occur at the subsurface of the contact zone, between rollers and raceways. Hence, it should be noted that some of the constraints related to strength appear in the constraint section.

In the above expressions λ is the reduction factor that accounts for manufacturing and mounting deviations (which is taken as 0.65), ν is the factor that accounts for the edge loading (which is taken as 1.2 for tapered roller bearings [18]). The factor b_m in equation (3) accommodates the improvements in bearing geometrical accuracies afforded by modern manufacturing methods and improvements in the modern rolling bearing steel chemistries and metallurgies. Values of b_m are as recommended by the bearing standard [4]. The numerical constant given in equation (4) accounts for properties of the bearing material used. In the present problem, material considered for the design is air-melted, high-chromium steel with grade SAE 52100, which has the property of high wear resistance. The objective function is related to the design parameters. For the present optimization problem, the objective function is maximization in nature; all the design variables are positive integers for this kind of problem. The following section gives details about the design variables' selection for the optimization problem.

3.2 Design Variables for the Optimization

From the observation of equations (3)-(5), the basic dynamic load capacity as a function of basic internal geometries of the bearing and unknown constraint constants (described in the subsequent section) may be expressed as

$$C_d = f(D_r, D_m, l_e, Z, K_{D\min}, K_{D\max}, \varepsilon, e, \beta) \quad (6)$$

Therefore, the four independent parameters D_r, D_m, l_e and Z have been selected as design variables. In addition to these, the five constraint constants ($K_{D\min}$, $K_{D\max}$, ε, e and β) are also taken as design variables, providing constraints to the basic design variables. Using trigonometric relations, all other internal geometries of the bearing can be related with the help of basic design variables and boundary dimensions. Therefore, the design variable vector can be expressed as

$$X = \{D_m, D_r, l_e, Z, K_{D\min}, K_{D\max}, \varepsilon, e, \beta\}^T \quad (7)$$

These design variables are subjected to some constraints, which are discussed in the subsequent sections.

3.3 Constraints Defining the Range of Geometric Design Variables

The present section describes the procedure for obtaining bounds of design variables, which define a more concise feasible search space for these variables, to be used in the genetic algorithm during optimization. These succinct bounds help in the faster convergence of the solution. It should be noted that these bounds are not arbitrary and are defined by using the mounting space available for a bearing and safe values of the contact stress and the subsurface shear stress. These bounds of design variables are fixed for a particular bearing number (for a given bearing material); however, they change with the bearing number (i.e., depend upon the bearing boundary to be optimized). Hence, these are some forms of constraints also.

Constraints 1 & 2

The pitch diameter of the bearing should be selected such that it lies between the inner and outer diameters of the bearing, ensuring the space for the chamfering of the sharp corners of the cup and the cone. Therefore, range of the pitch diameter can be given as

$$(d + 2r_{1_{\min}}) \le D_m \le (D - 2r_{3_{\min}}) \quad (8)$$

The corresponding constraint conditions can be given as

$$g_1(X) = D_m - (d + 2r_{1_{\min}}) \ge 0 \quad (9)$$

and

$$g_2(X) = (D - 2r_{3_{\min}}) - D_m \ge 0 \quad (10)$$

Constraints 3 & 4

The range of the mean diameter of the roller is derived from both the strength and geometric considerations. The lower limit of the roller mean diameter is obtained from the contact stress expression, and the upper limit is obtained from the geometry. The minimum value of the roller mean diameter should be such that it can withstand the contact stress developed due to external loads. The contact stress equation for the line contact [18] given as

$$\sigma^l_{c_{\max}} = \frac{2Q_{\max}}{\pi l_e b} \quad (11)$$

where

$$b = 3.35 \times 10^{-3}\left(\frac{Q_{\max}}{l_e \Sigma\rho}\right)^{1/2} \quad \text{and} \quad \Sigma\rho = \frac{2}{D_r(1-\gamma)} \approx \frac{2}{D_r} \quad (12)$$

where $\sigma^l_{c_{\max}}$ is the maximum contact stress, and $Q_{\max}$ is the contact load on the inner raceway at the most heavily loaded roller. In an equation of $\Sigma\rho$ the term $(1-\gamma)$ has been ignored, as it is nearly equal to 1. When substituting terms in equation

(12) into equation(11), we get

$$\sigma^{l}_{\max} = 268.71\sqrt{\frac{Q_{\max}}{D_r l_e}} \text{ where } D_r \approx 0.625 l_e \qquad (13)$$

To calculate the minimum value of D_r an approximate relation for D_r and l_e given in equation (13) [29] is used. From equation(13), the lower limit of roller mean diameter is obtained by taking the maximum contact stress as the safe contact stress (σ_{safe}). $Q_{\max}$ is obtained using Stribeck's conservative formula [18]. Maximum value of D_r is restricted by the boundary dimensions; i.e., D and d, and it can be obtained by considering the cup and cone to have their corners chamfered. The range for D_r can be taken as

$$D_{r_{L.L}} \leq D_r \leq D_{r_{U.L}} \qquad (14)$$

with

$$D_{r_{L.L}} = 212.43\frac{\sqrt{Q_{\max}}}{\sigma_{c_{\max}}} \quad \text{and}$$

$$D_{r_{U.L}} = \frac{1}{2}\{(D - 2r_{3_{\min}}) - (d + 2r_{1_{\min}})\} \qquad (15)$$

where $D_{r_{L.L}}$ is the lower limit of the mean diameter of roller and $D_{r_{U.L}}$ is the upper limit of the mean diameter of the roller. Equation (14) can be given in the form of constraints as

$$g_3(X) = D_r - 212.43\frac{\sqrt{Q_{\max}}}{\sigma_{safe}} \geq 0 \qquad (16)$$

and

$$g_4(X) = \frac{(D - 2r_{3_{\min}}) - (d + 2r_{1_{\min}})}{2\cos\alpha} - D_r \geq 0 \qquad (17)$$

Constraints 5 & 6

In roller bearings, the length of the roller is never less than its diameter. Moreover, the length of the roller should be such that it should not project from both the faces of the cup. Therefore, its range can be given as

$$(D_{r_{L.L}}) \leq l_e \leq l_{e_{U.L}} \qquad (18)$$

In the form of constraints, equation (18) can be written as

$$g_5(X) = l_e - (D_{r_{L.L}}) \geq 0 \qquad (19)$$

and

$$g_6(X) = \frac{(C - r_{5_{\min}} - r_{4_{\min}})}{\cos\alpha} - l_e \geq 0 \qquad (20)$$

Constraint 7 & 8

The minimum number of rollers is obtained based on the lower limit of the bearing pitch diameter and the upper limit of the roller diameter. Similarly, the maximum number of rollers is obtained based on the upper limit of the bearing pitch diameter and the lower limit of the roller diameter. The range for the number of rollers is given as

$$\frac{\pi(d + 2r_{1_{\min}})}{D_{r_{U.L}}} \leq Z \leq \frac{\pi(D - 2r_{3_{\min}})}{D_{r_{L.L}}} \qquad (21)$$

Constraints based on the range of number of rollers can be given as

$$g_7(X) = Z - \frac{\pi(d + 2r_{1_{\min}})}{D_{r_{U.L}}} \geq 0 \qquad (22)$$

and

$$g_8(X) = \frac{\pi(D - 2r_{3_{\min}})}{D_{r_{L.L}}} - Z \geq 0 \qquad (23)$$

3.4 Constraints for the Optimum Design of Tapered Roller Bearings

Constraints of basic design variables for the optimization problem are derived from the geometry, bearing standards, and strength considerations. In the present subsection, constraints for the optimum design of tapered roller bearings are summarized. These design constraints will helps to reduce the parameter space to the *feasible* parameter space for the optimization problem.

Constraints 9 & 10

The diameter of the rolling element should be chosen from certain bounds, that is

$$K_{D\min}\frac{D - d}{2\cos\alpha} \leq D_r \leq K_{D\max}\frac{D - d}{2\cos\alpha} \qquad (24)$$

where $K_{D\min}$ and $K_{D\max}$ are unknown constants (which decide, respectively, the possible minimum and maximum diameters of rolling element), and D and d are the outside and bore diameters of bearings, respectively. From equation (24), the corresponding constraint conditions are given as

$$g_9(X) = 2D_r - \frac{K_{D\min}(D - d)}{\cos\alpha} \geq 0 \qquad (25)$$

and

$$g_{10}(X) = \frac{K_{D\max}(D - d)}{\cos\alpha} - 2D_r \geq 0 \qquad (26)$$

In the present work, for the constraint constants (i.e., $K_{D\min}$ and $K_{D\max}$), instead of assigning a single arbitrary value to each of these constants, ranges have been chosen. They are based on the similar parametric study of Rao and Tiwari [31] on tapered roller bearings using the initial optimization runs. The ranges are chosen as $0.3 \leq K_{D\min} \leq 0.4$ and $0.5 \leq K_{D\max} \leq 0.6$.

Constraints 11 & 12

In order to guarantee the running mobility of bearings, the difference between the pitch diameter and the average diameter in a bearing should be less than a certain given value. Therefore, the following two constraints are to be satisfied.

$$g_{11}(X) = D_m - (0.5 - e)(D + d) \geq 0 \tag{27}$$

and

$$g_{12}(X) = (0.5 + e)(D + d) - D_m \geq 0 \tag{28}$$

where e is an unknown constraint constant, and the possible choices of it have been taken as $0.01 \leq e \leq 0.07$ using the parametric study [31] on tapered roller bearings.

Constraint 13

The thickness of bearing cup at the outer raceway bottom should not be less than εD_r, where ε is an unknown constraint constant. Therefore, the constraint condition is

$$g_{13}(X) = \frac{0.5(D - D_m - D_r)}{\cos\alpha} - \varepsilon D_r \geq 0 \tag{29}$$

The possible values of ε are chosen in the range of, $0.4 \leq \varepsilon \leq 0.5$ using the parametric study [31] on tapered roller bearings.

The following constraints are formulated based on internal geometric parameters obtained from the trigonometric relations. Using trigonometric relations, all other internal geometries of the bearing can be related with boundary dimensions and these relations include basic design variables (see Appendix B). Internal geometries are shown in Figure B.1 and are listed in the Nomenclature. These internal geometries locate the position and the orientation of the roller, and are expressed in terms of basic design variables and boundary dimensions of the bearing. The minimum width of the back-face of the cone-rib is designed based on its strength by using the maximum principal stress theory [24].

Constraint 14

From the fixed boundary dimensions of the bearing, the minimum thickness of the cup, $S^o_{2_{\min}}$, as shown in Figure B.1, can be expressed as [24]

$$g_{14}(X) = S^o_{2_{\min}} - \left\{\frac{1}{2}D - \left(\frac{1}{2}D_{oinner} + C\tan\alpha\right)\right\} \geq 0 \tag{30}$$

Constraint 15

In tapered roller bearings, the cone is designed to be stronger than the cup, since the cone gets more stresses because of its curvature. Therefore, the minimum thickness of the cone $S^i_{1_{\min}}$ should be larger than the minimum thickness of the cup, $S^o_{2_{\min}}$

$$g_{15}(X) = S^i_{1_{\min}} - S^o_{2_{\min}} \geq 0 \tag{31}$$

Constraint 16

To avoid the contact of the roller with the curved-end of the front-face of the cup and to avoid protrusion of the big-end of the roller, the minimum width of the front-face of the cup, $C_{2_{\min}}$, should be larger than the minimum value of r_5, as shown in Figure B.1. It also ensures that the corner of the big-end of the roller is away from the chamfering of the front-face of the cup; therefore the constraint on this condition can be given as

$$g_{16}(X) = C_{2_{\min}} - r_{5_{\min}} \geq 0 \tag{32}$$

Constrain 17

To avoid sharp corners of faces of the cup, they are chamfered and also the corners of the small end of the roller are kept away from the back-face chamfer of the cup. Hence, the constraint on the minimum width of the back-face of the cup, $C_{1_{\min}}$ (Figure B.1), can be given as

$$g_{17}(X) = C_{1_{\min}} - r_{4_{\min}} \geq 0 \tag{33}$$

Constraint 18

The corner of the big end of the roller should be away from the chamfer of the back-face of the cone; therefore the minimum width of the back-face of the cone, $B_{2_{\min}}$ (Figure B.1), has a constraint as

$$g_{18}(X) = B_{2_{\min}} - r_{2_{\min}} \geq 0 \tag{34}$$

Constraint 19

The corner of the small end of the roller should be away from the chamfer of the front-face of the cone; therefore the minimum width of the front-face of the cone, $B_{1_{\min}}$ (Figure B.1), has a constraint as

$$g_{19}(X) = B_{1_{\min}} - r_{5_{\min}} \geq 0 \tag{35}$$

Constraint 20

The larger lip of the cone is the critical element that takes the axial component of forces, whereas the smaller lip of the cone retains the rollers on the cone. Therefore, the larger lip of the cone should be stronger than the smaller lip, which gives the constraint as

$$g_{20}(X) = B_{2_{\min}} - B_{1_{\min}} \geq 0 \tag{36}$$

Constraint 21

In tapered roller bearings, the width of the back-face of the cup should be larger than the width of the front-face of the cup. Therefore the constraint can be given as

$$g_{21}(X) = C_{1_{\min}} - C_{2_{\min}} \geq 0 \tag{37}$$

Constraint 22

The constraint on the length of the roller is a critical concern in the design of roller bearings. The effective length, l_e, and the width of the bearing cup, C, gives constraint on the maximum length of the roller (i.e., $l_e < \beta C$) and the constraint can be written as

$$g_{22}(X) = \frac{\beta C}{\cos\alpha} - l_e \geq 0 \tag{38}$$

where β is an unknown constraint constant, and has been chosen in the range of $0.8 \leq \beta \leq 0.95$ using a parametric study [31] on tapered roller bearings.

Constraint 23

The small end of the roller should not protrude from the cup; the constraint to fulfill this condition can be given as

$$g_{23}(X) = \left(\frac{1}{2}D - S^o_{1_{\min}}\right) - \frac{1}{2}D_{oinner} \geq 0 \tag{39}$$

Constraint 24

The maximum principal stress at the flange of the cone due to the flange load should be within the yield strength of the material of the flange. The constraint can be given as

$$g_{24}(X) = \sigma_y - \sigma_{f_{\max}} \geq 0 \tag{40}$$

where σ_y is the yield strength of material of the flange, and $\sigma_{f_{\max}}$ is the maximum principal stress in the flange. Detailed discussion about all these stresses in the flange is given in Appendix C.

Constraint 25

In rolling bearings large amounts of force are concentrated at a relatively small contact area, resulting in very high values of contact stresses, which leads to some degree of permanent deformation at contact surfaces [18]. However, experience has shown that for the safe operation of the bearing the permanent deformation should be within 0.0001 times the mean diameter of the rolling element diameter when the bearing is loaded with an equivalent radial equals to the dynamic load capacity. Alternatively, in terms of the contact stress, the maximum contact stress should not exceed 4000N/mm^2. Thus,

$$g_{25}(X) = \sigma_{safe} - \sigma^l_{\max} \geq 0 \tag{41}$$

where σ_{safe} is the safe contact stress, $\sigma^l_{\max}$ is the actual contact stress at the line contact between the cone and the roller when the roller is loaded with $Q_{\max_i}$ over the cone. The contact stress equation for the line contact (Hertzian contact) is taken from [18].

Constraint 26

In the tapered roller bearings, rollers roll on the cup and the cone. To ensure smooth rolling of rollers and to avoid collision between them, proper spacing should be provided between the rollers. A minimum angular spacing of one degree is to be allowed as in standard bearings. This constraint will limit the number of rollers, which can be accommodated in a bearing. Thus,

$$g_{26}(X) = 2\pi - 2Z\sin^{-1}\left(\frac{D_r\cos\alpha}{D_m}\right) \geq Z\frac{\pi}{180} \tag{42}$$

4. COMPUTATIONAL PROCEDURES

The computational technique used for solving the constrained optimization problem that is formulated in previous sections is real-coded GAs. It mimics the principles of natural genetics to constitute the search and optimization procedures. The function of GA begins with the selection of GA parameters such as population size, selection operator, mutation probability, crossover probability, crossover distribution index, and mutation distribution index. Design variables are generated within the range according to the formulation discussed in section 3.3 using a random number. The fitness function (i.e., maximization of the basic dynamic load capacity) is evaluated. The fitness function basically determines which possible solutions get passed on to multiply and mutate into the next generation of solutions. This is usually done by analyzing the "genes," which hold some data about a particular solution to the problem that it is trying to solve. The fitness function will look at the genes and make some qualitative assessment, returning a fitness value for that solution. The rest of the genetic algorithm will discard any solutions with a "poor" fitness value and accept any with a "good" fitness value. The optimization techniques search for the feasible solution in the solution space. The design point normally represents the global optimum, owing to the evolutionary nature of the algorithm. Computations were conducted on a Pentium-4 HT 3.2 GHz machine with 1 GB of RAM using Microsoft-Visual C++6.0.

The present optimization problem is a nonlinear constrained optimization problem (formulated in Sections 3.3 and 3.4). In trying to solve the constrained optimization problems using genetic algorithms (GAs), the selection of constraint handling methods plays an important role. There are several ways to handle constraints in optimization problems. Both the penalty function [19] and penalty parameter-less [12] approaches have been most popularly applied with GA techniques because of their simplicity and ease of implementation. For the present optimization problem, the penalty parameter-less approach is used to handle the constraints and this approach is best suited for optimization algorithms working with the population.

4.1 Real Coded Genetic Algorithm

Over the past few years, many researchers have been paying attention to real-coded evolutionary algorithms, particularly for solving real world optimization problems. Standard genetic algorithms use the binary coding of the variables (i.e., variables are represented as strings of 0s and 1s). Working with the binary coding of variables converts the search space into a discrete set of points, even though the underlying objective function is a

TABLE 1
Input parameters used for the optimal design of tapered roller bearings

	Standard boundary dimensions					Standard internal dimensions		Standard chamfering dimensions						Dynamic load rating,
Bearing No.	D mm	d mm	C mm	B mm	T mm	d_1 mm	D_{oinner} mm	α^o	$r_{1_{\min}}$ mm	$r_{2_{\min}}$ mm	$r_{3_{\min}}$ mm	$r_{4_{\min}}$ mm	$r_{5_{\min}}$ mm	C_{ds} kN
30204	47	20	12	14	15.25	33.20	37.304	12.9527	1.0	1.0	1.0	1.0	0.5	27.50
30205	52	25	13	15	16.25	37.40	41.135	14.0361	1.0	1.0	1.0	1.0	0.5	30.80
32205	52	25	15	18	19.25	40.20	37.555	21.2500	1.0	1.0	1.0	1.0	0.5	35.80
322/28	58	28	16	19	20.25	43.90	42.436	20.5666	1.0	1.0	1.0	1.0	0.5	41.80
32206	62	30	17	20	21.25	45.20	48.982	14.0361	1.0	1.0	1.0	1.0	0.5	50.10
30207	72	35	15	17	18.25	51.80	58.844	14.0361	1.5	1.5	1.5	1.5	0.5	51.20
30306	72	30	16	19	20.75	48.40	58.287	11.8597	1.5	1.5	1.5	1.5	0.4	56.10
32207	72	35	19	23	24.25	52.40	57.087	14.0361	1.5	1.5	1.5	1.5	0.5	66.00
30307	80	35	18	21	22.75	54.50	65.769	11.8597	2.0	2.0	1.5	1.5	0.8	72.10
32208	80	40	19	23	24.75	58.40	64.715	14.0361	1.5	1.5	1.5	1.5	0.5	74.80

continuous function. In order to obtain the optimum point with a desired accuracy, strings of sufficient length are to be chosen. GAs have also been developed to work directly with continuous variables where variables are directly used without coding, known as the real coding. In the present optimization problem, the real coding of variables is used [10]. A single-objective, real-coded genetic algorithm is applied to the present optimization problem. Figure 3 illustrates the flow chart for the present optimization problem and describes the step-by-step procedure for solving the given optimization problem. A real-coded genetic algorithm has many advantages over binary-coded genetic algorithms [21].

GA consists of three major operators: reproduction, crossover, and mutation. These operators are mainly applied to update the solution in every generation. The detailed explanations of these operators are given in [21]. In the present case, real-coded GAs are used with the simulated binary crossover (SBX) operator and a parameter-based mutation or polynomial mutation operator [13]. The SBX operator is particularly suitable, because the spread of children solution around the parent solution can be controlled using a crossover distribution index η_c and the diversity of parent solution can be controlled using a mutation distribution index η_m.

TABLE 2
Material properties of the bearing steel

S. No	Description	Values
1	Safe contact stress	4000 MPa
2	Young's modulus	210 GPa
3	Yield strength	600 MPa
4	Poisson's ratio	0.3

5. DESIGN APPLICATIONS AND RESULTS

A number of tapered roller bearings were selected from bearing standards, belonging to various dimensional and contact angle series for obtaining the optimal design by using the proposed methodology. Table 1 provides the standard boundary, internal and chamfering dimensions, which are taken from the SKF general catalogue [32] and the bearing standard [3]. The table also lists the standard C_d of the bearing from the SKF general catalogue [32], from which 60% of C_d is taken as an equivalent external radial load applied to the bearing for the design of bearing components based on geometric and strength considerations. Table 2 lists the material properties to be used in the strength constraints. All the data listed in Tables 1 and 2 would be used as an input for the proposed optimum design methodology of tapered roller bearings. In addition to standard bearing parameters, a few GA parameters are also taken as input, which are listed in Table 3.

5.1. GA Implementation and Numerical Illustrations

Tapered roller bearings mentioned in Table 1 have been optimized serially. For a particular bearing number, first bounds of

TABLE 3
Parameters chosen for genetic algorithms

Parameter	Value
Crossover probability	0.9
Mutation probability	0.2
Population size	200
Number of generations	100
Index of crossover	50
Index of mutation	20
Tournament size	2

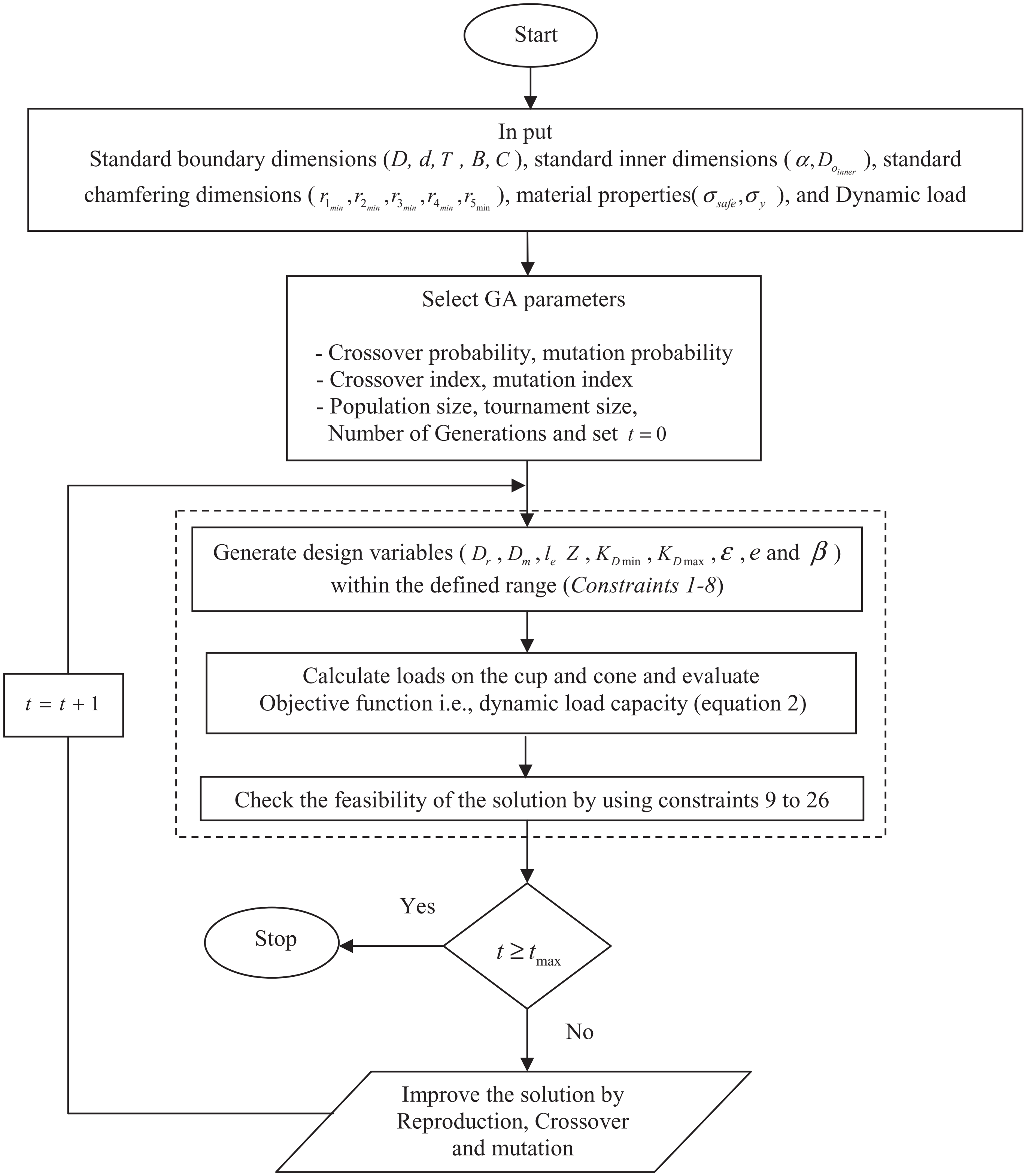

FIG. 3. A flow diagram of the optimum design of tapered roller bearings.

TABLE 4
Bounds of basic design variables for 30204

Design variable	Lower limit	Upper limit
D_m (mm)	22.00	45.00
D_r (mm)	1.5454	11.8003
l_e	1.5454	10.7741
Z	5	91
$K_{D\,\min}$	0.30	0.40
$K_{D\,\max}$	0.50	0.60
e	0.40	0.50
ε	0.01	0.07
β	0.80	0.95

basic design variables are obtained from constraints 1 to 8. The lower and upper bounds of basic design variables are obtained by using the standard boundary dimension and other geometries specified. The expression for contact stress as given by equation (15) requires the maximum contact load to be calculated by using Stribeck's formula [18]. This formula contains a radial load to be applied to the bearing and number of rollers in the bearing. Using 60% of C_d given in the standard SKF catalogue as the equivalent radial load and assuming an arbitrary high number of rollers equal to 100, the lower limit of the diameter of the roller is computed. The bearing model 30204 has been taken for the study of convergence. Input parameters for optimizing the design of bearing model 30204 are listed in Table 1. Bounds of basic design variables obtained from the procedure described above are listed in Table 4. It should be noted that, by providing bounds, the search space becomes narrowed and it leads to faster convergence. For the present optimization problem, the crossover probability of 0.9 and the mutation probability of 0.2 have been taken for all the sequence of bearing numbers given in Table 1. Distribution indexes for the crossover and the mutation are taken as 50 and 20, respectively.

Using bounds listed in Table 4, basic design variables are generated for a population size of 200. The algorithm is run with a certain number of generations and it is increased in succession until the optimum point in the design remains unchanged with further increase in generations, which indicates that the convergence has achieved. Based on the convergence point in the present optimization problem, the number of generations has been taken as 100. The scatter of fitness function value of the whole population for the initial generation and for the last generation is shown in Figure 4. It includes all the population, including solutions violating the constraints, and from the choice of the initial solutions it can be seen that it has quite good coverage of the possible solution domain. In the last generation all the solutions have also been shown, including those slightly violating constraints, and it could be observed that most of the solutions converged to around 31.220 kN.

Figure 5 shows the relation between average fitness function value and the number of generations. During initial generations,

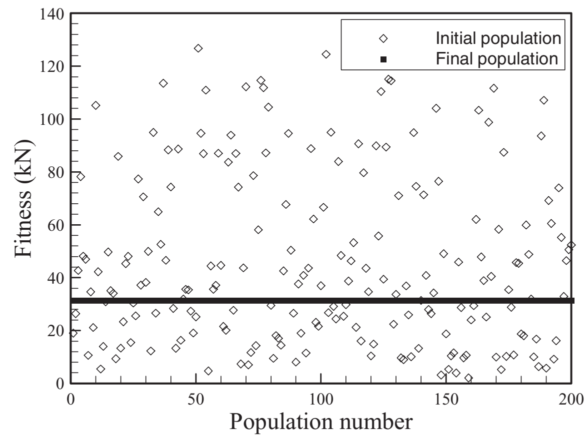

FIG. 4. Variation of C_d with population number for bearing 30204.

the average fitness values are high; however, as the generation number increases the average fitness value decreases up to the value of 17.49 kN, and thereafter the fitness value again increases in its magnitude (i.e., beyond 8 generations) up to generation number 23 in a drastic manner. After that, the fitness values continuously increase up to convergence point (i.e., 70 generations). After that, the fitness value seems to more or less stable in its magnitude, having a fitness value of 30.74 kN. The average fitness is obtained by dividing the sum of fitness values in a particular generation with the population size. The convergence study of average fitness function value shows that the convergence is fast. Figure 6 shows the variation of best solution with the number of generations and this is obtained by choosing the best solution in each generation. At initial generations the solution is converged to a smaller value of fitness (i.e., 15.22 kN), which is up to the sixth generation; from generation number 7, a sudden change in fitness value took place. Beyond that, fitness value was stabilized to 17.62 kN. Then after, again a sudden improvement in the fitness function value was observed

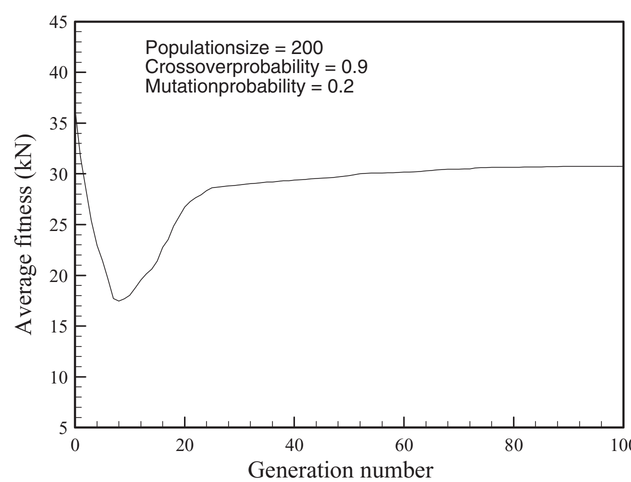

FIG. 5. Variation of the average fitness with number of generations for bearing 30204.

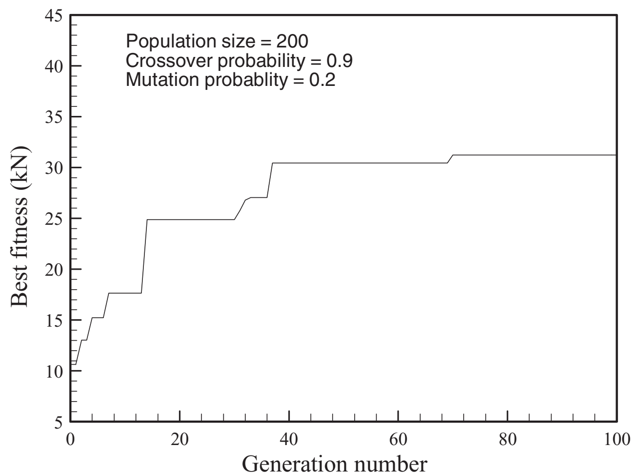

FIG. 6. Best fitness vs number of generations for bearing 30204.

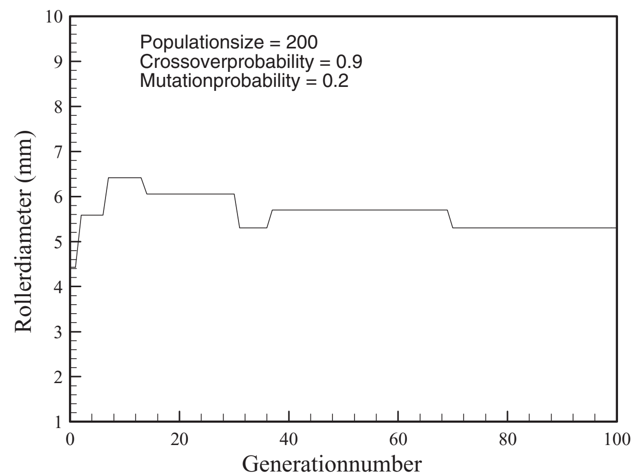

FIG. 8. Roller diameter vs number of generations for bearing 30204.

at generation number 14, and then it stabilized at a fitness value of 24.86 kN. For the rest of the generations, a process of improvement and stabilization of fitness values also took place. At generation numbers 31, 37, and beyond 70, the respective fitness values of 25.76, 30.43, and 31.22 kN were observed. However, in between these generations (31, 37, and 70) and beyond generation number 70, the fitness value remains stable. From Figure 6, a timely improvement of the best solution along the number of generations has been observed clearly.

Figure 7 shows the variation of bearing pitch diameter of the best solution with the number of generations. At initial generations, the bearing pitch diameter is high (i.e., 35.823mm); however, at the convergence point (i.e., at generation number 70), the pitch diameter value of 34.999 mm is observed. Beyond that, the pitch diameter value remains the same for the rest of the generations. Figure 8 shows the variation of roller diameter of the best solution with the number of generations. Initially, roller diameter values fluctuate along with generations. The convergence of the roller diameter with a value of 5.299 mm has been observed at the 70th generation; beyond that, the value is the same for the next generations. Figure 9 shows the variation of effective length of the roller of the best solution in each generation with the generation number. At the 70th generation, the effective length converged to a value of 10 mm; beyond the 70th generation the value was the same. Figure 10 shows the variation in number of rollers of the best solution with the number of generations. Finally, at generation number 70, the number of rollers reaches 20; the value remains the same for the rest of the generations. The convergence study shows that the convergence of design parameters is fast in nature. Figures 7, 8, 9, and 10 show the timely convergence of the solution along with the number of generations for all the design parameters. Those plots, which show the timely convergence, can give information regarding optimal solution variation according to the generation number.

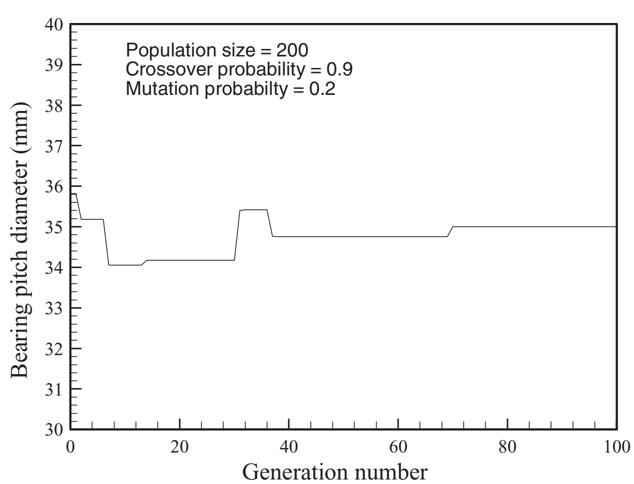

FIG. 7. Bearing pitch diameter vs number of generations for bearing 30204.

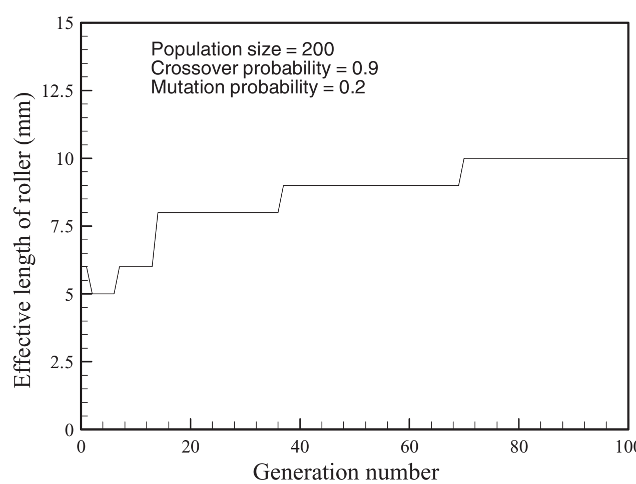

FIG. 9. Effective length of roller vs number of generations bearing 30204.

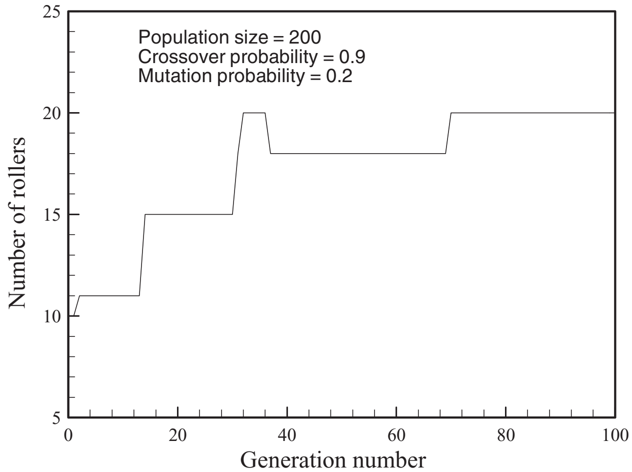

FIG. 10. Number of rollers vs number of generations for bearing 30204.

For every stochastic optimization problem, there are two important qualities: the solution quality and time requirements. There are a number of other qualities that also might be of interest, namely the flexibility, understandability, and ease of implementation. Normally, the solution quality is measured in terms of percentage above the standard solution (e.g., from [32]) for maximization problem, although for some applications an absolute difference is the desired measure. From the optimized values, it is observed that a 13.52% increment in the value of C_d compared to standard C_d value for a bearing number of 30204. The bearing pitch diameter is 2.29% higher than the standard value can be observed. Figure 11 represents the comparison of optimized design parameters and fatigue life normalized by those of the standard design and with standard life (i.e., from C_{d_s}), respectively. From Figure 11, it can be seen that the optimized design parameters D_m, l_e and Z increase, and D_r decreases. The L_{10} life of the bearing number 30204 is increased by 13.52 percent. Maximum computational time for the algorithm (for a single-bearing configuration) is 20 sec. A drawing of a typical optimized roller bearing design indicating optimized values of design parameters for the tapered roller bearing number 30204 is given in Figure 12. The values of maximum contact load, contact stress, depth of maximum dynamic shear in the cone, and maximum dynamic shear stress for the typical bearing 30204 are listed in Table 5.

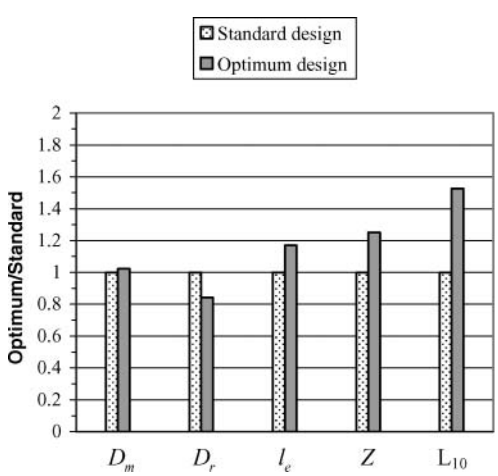

FIG. 11. Comparison of the optimized design parameters and fatigue life with standard design.

TABLE 5
Contact and shear stresses for tapered roller bearing 30204

S.N.	Description	Values
1	Maximum contact load on the cone	4453.20 N
2	Maximum contact stress on the cone	2656.96 MPa
3	Depth of maximum dynamic shear in the cone	0.0667 mm
4	Maximum dynamic shear in the cone	1328.48 MPa
5	Maximum principal stress in the flange	600 N/mm^2

The optimized internal geometries of bearings are listed in Table 6, which include the optimized four basic geometrical design variables and five design constraint constants. It also includes eight post-processed internal geometric variables of bearing from the optimized basic geometric design variables, which are obtained by taking input from the standard geometrical data given in Table 1. It could be observed that as the size of the bearing increases the pitch diameter has a tendency to increase in the same proportion; it is expected due to some of the constraints (e.g., 1, 2, 11, and 12). The roller diameter also has somewhat similar trends. The increment in the effective length of roller has been observed for some of the bearing numbers corresponding to the standard boundary dimensions, which are tabulated in Table 1. Some of the bearing numbers have the same converged value of effective length, even though their boundary

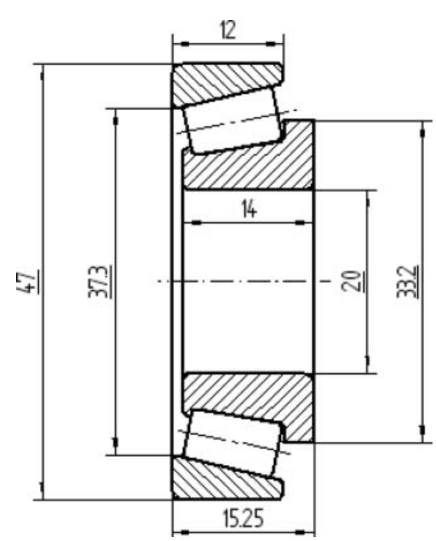

FIG. 12. Optimum design drawing of a typical tapered roller bearing 30204 (all dimensions in mm).

dimensions are not the same. The number of rollers are found to change in the magnitude with the bearing number (with the maximum change of five rollers); however, it does not take a very high number of rollers (i.e., of the order of 91 as in Table 4). For some of the bearing numbers, the converged number of rollers are found to be the same in magnitude, even though they have different boundary dimensions. Hence, it is not always that the larger number of rollers is always advantageous, even though a larger diameter of rollers with a certain minimum number of rollers could also provide better fatigue life. It could be due to the fact that the large number of rollers would provide the higher frequency of the cyclic load (it might be of smaller magnitude). and this would lead to the lesser fatigue life. From these discussions, it could be concluded that it is difficult to frame a general rule that a particular design variable has more effect on the fatigue life as compared to the others; however, it depends upon different bearings and operating parameters.

Constraint constants have been found to converge most of the time at all of the chosen range for different bearings, and sometimes at the boundary of bounds also. These constants have no definite trend after the convergence. For example, for the case of bearing number 30204 all converged constants are well within the boundaries of chosen bounds (Table 6). By observing the optimized design constraint constant values from Table 6, it is concluded that for most of the bearing numbers, the constraint constant $K_{D\min}$ is converged near to its lower bound values. The constraint constant $K_{D\max}$ is converged near to its upper bound values for many of the bearing numbers. The constraint constantε, which accounts for outer ring strength consideration, has been optimized near to the upper bound values for most of the bearing numbers. The constraint constant e, which accounts for the bearing mobility condition, in most of the cases converges near to its upper limit value. The constraint constant β, which accounts for maximum length of the roller, has been optimized most of the time to its upper bound value. The optimized values of the above design constraint constants for each bearing have been tabulated serially in Table 6. To compare the life of the optimum designed bearing, the following equation is used

$$\lambda_l = \frac{L_g}{L_s} = \left(\frac{C_{d_g}}{C_{d_s}}\right)^{10/3} \tag{43}$$

Equation (43) is obtained from equation (4) for the same value of the applied equivalent radial load. The subscripts "g" and "s" represent the values obtained by using the GA and standard catalogue, respectively. The bearing life comparison parameter (i.e., λ_l values) is tabulated serially in Table 6 for all bearing numbers considered. On an average, the life obtained by the optimum design values is more or less 1.5 times better than the existing bearing catalogue values (i.e., SKF, 2005). In Table 7, a comparison of optimized bearing geometry with the standard geometry for a tapered roller bearing of 30204 is made.

5.2. Parametric Sensitivity Analyses Using Monte Carlo Simulation Technique

At the time of manufacturing, it is not uncommon to see 0.5-1% deviation in the design parameter values. To better evaluate a bearing design, it is important to observe the sensitivity in the optimized values of the obtained performance measures with respect to such parametric variations. The sensitivity analysis is used to quantify the impact of changes in the initial data of nonlinear programs on optimal solutions. In particular, the parametric sensitivity analysis involves a perturbation analysis in which the effects of small changes of some or all of the initial data on an optimal solution are investigated, and the optimal solution is studied on a so-called critical range of the initial data, in which certain properties such as the optimal basis in the nonlinear programming are not changed. This section attempts to address this concern of designs.

The Monte Carlo simulation sensitivity analysis technique has been used for the present optimization problem for a robustness check due to manufacturing tolerances. Monte Carlo simulation techniques work based on the probabilistic approach and rely on repeated random samplings to compute the results. The technique was first used by von Neumann [22] during World War II as a code word for secret work on nuclear weapons at Los Alamos Laboratory in New Mexico. This technique has evolved as a very powerful tool for engineers with only a basic working knowledge of probability and statistics for evaluating the risk or reliability of complicated engineering systems. Some of the advantages of Monte Carlo sensitivity analysis techniques are: (i) for optimization problems, Monte Carlo simulation often can reach global optimum and overcome local extreme; (ii) to use Monte Carlo simulation is quite simple and straightforward, as long as the convergence can be guaranteed by the theory; (iii) the method can be used for both stochastic (involve probability) and deterministic (without probability); and (iv) Monte Carlo simulation is one of the largest and most important classes of numerical methods for computer simulations.

The Monte Carlo simulation technique consists of six essential elements: (1) select the parameters to be tested for sensitivity; (2) set the range of each parameter to include the variations experienced at the time of manufacturing; (3) generate the values of these random variables, for example N independent random numbers with a uniform distribution within their ranges; (4) evaluate the problem deterministically for each set of realizations of all the random variables (i.e. calculate the objective function values); (5) extract the probabilistic information from N such realizations; (6) determine the accuracy and efficiency of the simulation [9].

Out of all the parameters detailed in section 3.2, the ones that influence the performance measures and can suffer from manufacturing tolerances are D_m, D_r, and l_e. Therefore, we will study the performance impact of varying these parameters by a maximum of $\pm 1\%$ around a given optimized solution point. Variation in parameters has been done by generating random points within a $\pm 1\%$ neighborhood of the corresponding value. A

TABLE 6

Optimized design parameters, constraint constants, and obtained internal geometric variables for tapered roller bearings through GA

Bearing No	Optimized basic design variables				Optimized design constraint constants					Obtained internal geometric variables of bearing from the optimized basic design variables									
	D_m mm	D_r mm	l_e mm	Z	$K_{D\min}$	$K_{D\max}$	ε	e	β	β^o	$S^o_{2_{\min}}$ mm	$S^i_{1_{\min}}$ mm	$B_{1_{\min}}$ mm	$B_{2_{\min}}$ mm	$C_{1_{\min}}$ mm	$C_{2_{\min}}$ mm	α^o_f	C_{d_g} kN	λ_l
30204	34.999	5.29	10.0	20	0.3306	0.5847	0.4507	0.0623	0.9465	1.693	2.437	3.665	1.509	3.603	1.703	1.520	9.757	31.22	1.526
30205	39.366	5.38	11.0	21	0.3011	0.5479	0.4658	0.0575	0.9447	1.676	2.352	3.534	1.475	2.710	1.642	0.681	10.79	34.81	1.503
32205	38.646	5.38	13.0	22	0.3278	0.5978	0.4464	0.0517	0.9494	2.558	1.766	2.463	2.192	3.307	1.901	0.969	16.71	40.60	1.521
322/28	42.839	6.14	14.0	21	0.3260	0.5987	0.4683	0.0699	0.9499	2.541	2.197	2.627	2.222	3.272	1.763	1.116	16.00	48.54	1.645
32206	47.329	6.37	15.0	21	0.3647	0.5993	0.4983	0.0666	0.9495	1.654	2.401	4.154	1.897	3.358	1.873	0.568	10.80	52.25	1.150
30207	56.117	6.89	13.0	23	0.3305	0.5027	0.4666	0.0699	0.9417	1.524	3.000	5.955	1.863	2.370	1.695	0.688	10.90	54.51	1.232
30306	54.221	7.74	14.0	20	0.3451	0.5597	0.4999	0.0699	0.9498	1.474	3.643	7.219	1.175	3.988	1.595	0.698	8.797	59.35	1.206
32207	54.381	7.99	16.0	20	0.3516	0.5968	0.4678	0.0678	0.9204	1.786	2.962	4.331	2.787	4.471	2.447	1.023	10.69	69.81	1.205
30307	60.875	8.77	15.0	20	0.3545	0.5823	0.4853	0.0643	0.9043	1.531	3.556	7.325	2.069	4.101	2.264	1.050	8.739	74.94	1.137
32208	60.800	9.06	15.0	20	0.3847	0.5988	0.4517	0.0390	0.8626	1.809	3.348	4.611	2.685	5.554	2.615	1.825	10.70	75.42	1.027

TABLE 7
Comparison of optimized bearing geometry with standard bearing geometry for 30204

S. No	Bearing parameters	Standard bearing dimensions for 30204*	Obtained and optimized bearing dimensions for 30204	Percentage change in the parameters	Remarks (Increases: (↑), decreases: (↓))
1	D_m (mm)	34.214	34.999	2.294	(↑)
2	D_r (mm)	6.29	5.29	15.898	(↓)
3	$D_{r_{\max}}$ (mm)	6.634	5.609	15.450	(↓)
4	$D_{r_{\min}}$ (mm)	5.946	4.970	16.414	(↓)
5	l (mm)	9.9	10.80	9.090	(↑)
6	l_e (mm)	8.55	10.0	16.959	(↑)
7	β^o(degree)	2.043	1.693	17.131	(↓)
8	S_i (mm)	3.176	3.665	15.396	(↑)
9	S_o (mm)	2.196	2.437	10.974	(↑)
10	$C_{1_{\min}}$ (mm)	1.4169	1.703	20.191	(↑)
11	$C_{2_{\min}}$ (mm)	0.9288	1.520	63.652	(↑)
12	$B_{1_{\min}}$ (mm)	1.29	1.509	16.976	(↑)
13	$B_{2_{\min}}$ (mm)	2.9215	3.603	23.327	(↑)
14	Z	16	20	25.00	(↑)
15	α_f (degree)	9.53	9.757	2.381	(↑)
16	C_d (kN)	27.5	31.220	13.527	(↑)

* SKF

uniform distribution was used to generate these random points. For instance, when doing the sensitivity analysis with respect to one of the design parameters, like the pitch diameterD_m, it is varied by ±1% around its final converged value and all other parameters are kept fixed. We will demonstrate the sensitivity of the objective function, i.e., C_d, with respect to each of these parameters separately and collectively. Figure 13 represents the flow chart for the modeling procedure of the sensitivity of single- and multi-parametric analyses.

The sensitive analysis is applied for 30204 bearing number. It has been observed that there is not much variation in the C_d, ±1% deviation of the bearing pitch diameter; the overall variation in the C_d is even less than 0.145%. The C_d decreases up to 1.145% by 1% decrement of the roller diameter and increases up to 1.092% by 1% increment in the roller diameter. Similarly, C_d decreases by 0.777% with 1% decrement in the effective length of the roller and increases by 0.776% with 1% increment in the effective length of the roller. In the above analysis, we discussed the effect on objective function when only the single parameter variation was done. The simultaneous variations in the design parameters are discussed below. The variation in C_d, by ±1% simultaneous variation in the design parameters D_m and D_r, is given in Table 8. The value of C_d decreases by 1.15% due to 1% decrement in D_m and D_r, whereas there is a 1.12% increment observed in C_d due to 1% increase in D_m and D_r. From the results, there is 0.826% decrement and 0.844% increment in C_d values observed due to 1% decrease and 1% increase in D_m and l_e, respectively. The percentage variations in C_d for ±1% deviation in parameters D_r and l_e are given. The variation in C_d is

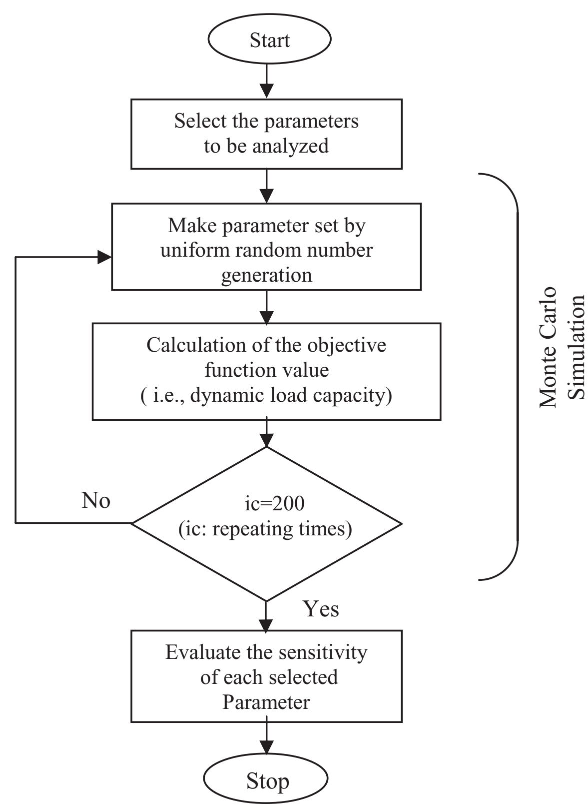

FIG. 13. A flow diagram of the Monte Carlo simulation for the sensitivity analysis.

TABLE 8
Parametric sensitivity analysis of the optimized design parameters for 30204

+1% deviation of optimized design variables			% change in objective function value	−1% deviation of optimized design variables			% change in objective function value
D_m (mm)	D_r (mm)	l_e (mm)	(C_d)	D_m (mm)	D_r (mm)	l_e (mm)	(C_d)
√	×	×	0.0703	√	×	×	−0.0717
×	√	×	1.0927	×	√	×	−1.1444
×	×	√	0.7765	×	×	√	−0.7779
√	√	×	1.1227	√	√	×	−1.1494
√	×	√	0.8446	√	×	√	−0.8264
×	√	√	1.7840	×	√	√	−1.7367
√	√	√	1.8232	√	√	√	−1.7416

large compared to variation in other parameters, where 1.736% decrement and 1.784% increment in C_d values was observed due to 1% decrement and 1% increments in D_r and l_e, respectively. When all the aforementioned parameters (D_m, D_r and l_e) were varied simultaneously, a trend similar to a collective variation in D_r and l_e was found. The parametric sensitivity values of the optimized design parameters for 30204 bearing are tabulated in Table 8. In all the discussed cases, the constraint violations have been checked and found to be all right. From the above observations, the percentage variations of C_d due to the sensitivity of design parameters are very small in magnitude. Due to more design parameters, variations could be expected from the objective function and/or constraints violations.

The objective function is found to be well-behaved during this parametric sensitivity analysis. However, in spite of the large number of constraints, they were found not to violate during the sensitivity analysis. This is because of the small variations in objective function C_d by ±1% variation in optimized design geometric parameters. The present optimization problem is well constrained with realistic constraints; because of this there is less chance of variation of C_d. The realistic constraints that contain the five design constraint constants are always try to converge quickly to the solution near the optimal point. For the same percentage variations in design parameters, the variation in C_d is less in tapered roller bearings compared to ball bearings [17]. This is due to the fact that the objective function formula of the tapered roller bearing is different from the ball bearing, even though the objective (i.e., maximization of C_d) is the same for both cases. In the case of ball bearings, the inner raceway groove curvature is found to be critical; a slight change in its value changes C_d drastically. However, for taper roller bearings this particular geometrical parameter does not exist. From the sensitivity analysis, it could be concluded that for tapered roller bearings most of the optimized variables have little effect on the performance parameter around its optimized values. It demonstrates that the solutions obtained are robust against marginal parametric variations that are hard to avoid during the time of manufacturing.

6. CONCLUSIONS

In the present paper, a procedure for the optimum design of tapered roller bearings has been proposed. The design problem has been optimized using real-coded genetic algorithms. The maximum fatigue life of the bearing is taken as the objective function and is subjected to nonlinear constraints. Constraints are formulated mainly based on geometrical and strength considerations. Design parameters selected for the optimization problem include the nominal diameter of roller, pitch diameter of the bearing, effective length of the roller, number of rolling elements, and the five design constraint constant parameters ($K_{D\,\min}$, $K_{D\,\max}$, ε, e and β), which define the complete basic inner geometry of the bearing and are also necessary for calculating the objective function C_d. Range for the design variables has been derived in terms of standard basic boundary dimensions, and it is observed that the design parameters and design constraint constants are converged to a very narrow range. A convergence study has been carried out to ensure the global optimum point in the design. There is good agreement between the optimized and standard bearings in respect of the C_d. A parametric sensitivity analysis was also carried out to understand the effect of deviation of design variables due to manufacturing tolerances, and the results show that no geometric design parameters have an adverse effect on C_d. The proposed optimum design methodology could be considered as a basic step towards a more advance design requirement of the tapered roller bearing, which requires a multitude of factors such as fits and tolerances, diametral clearance, complex static and dynamic loadings with associated contact stress analysis, heat generation, elastohydrodynamic lubrication, crowning of contacting surfaces (rollers, races), etc.

REFERENCES

1. S. Andreason, Load Distribution in a Tapered Roller Bearing Arrangement Considering Misalignment, Journal of Tribology, vol. 2, pp. 84–92, 1973.
2. I. Bercea and S. Cretu, Analysis of Double-row Tapered Roller Bearings, Part I, Tribology Transactions, vol. 46, pp. 228–239, 2003.

3. IS 7461, Part 1): *General Plan of Boundary Dimensions for Tapered Roller Bearings*, Bureau of Indian Standards, New Delhi, India, 1993.
4. IS 3824, *Rolling Bearings: Dynamic Load Ratings and Rating Life*, Bureau of Indian Standards, New Delhi, India, 2003.
5. W. Changsen, *Analysis of Rolling Element Bearings*, Mechanical Engineering Publications Ltd., London, 1991.
6. I. Chakraborty, V. Kumar, S. B. Nair, and R. Tiwari, Rolling Element Bearing Design through Genetic Algorithm, Engineering Optimization, vol. 35(6), pp. 649–659, 2003.
7. N. Chaturbhuj, S. B. Nair, and R. Tiwari, Design Optimization for Tapered Roller Bearings Using Genetic Algorithm, *Proceedings of the International Conference on Artificial Intelligence, IC-AI'03, June 23-26*, 2003, Las Vegas, Nevada, USA, Volume 1, Athens, Greece, CSREA Press, 2003, pp. 421–427.
8. D. Chaturvedi, K. Deb, and S. K. Chakrabarty, Structural Optimization Using Real-coded Genetic Algorithms, *Proceedings of the Symposium on Genetic Algorithms, March 1995*, Dehradun, India, 1995, pp. 73–82.
9. J. Choi, W. J. Harvey, and H. M. Concklin, Use of Multi-parameter Sensitivity Analysis to Determine Relative Importance of Factors Influencing Natural Attenuation of Mining Contaminants, Journal of Hydrology, vol. 209, pp. 297–304, 1998.
10. K. Deb, Optimization for Engineering Design, *Algorithms and Examples*, Prentice-Hall of India Private Limited, New Delhi, 1996.
11. K. Deb and S. Agrawal, A Niched-penalty Approach for Constraint Handling in Genetic Algorithms, *Proceedings of the International Conference on Artificial Neural Networks and Genetic Algorithms*, 1999, pp. 235–243.
12. K. Deb, An Efficient Constraint Handling Method for Genetic Algorithms, *Computer Methods in Applied Mechanics and Engineering*, vol. 186, pp. 311–338, 2000.
13. K. Deb, *Multi Objective Optimization for Engineering Design Algorithms*, Prentice-Hall of India, New Delhi, 2003.
14. K. Deb and S. Tiwari, Omni-optimizer: A Generic Evolutionary Algorithm for Single and Multi-objective Optimization, European Journal of Operation Research, vol. 185, pp. 1062–1087, 2008.
15. L. Eshelman and J. D. Schaffer, Real-coded Genetic Algorithms and Interval-schemata, *Foundations of Genetic Algorithms II*, Morgan Kaufmann, San Mateo, CA, 1993, pp. 187–202.
16. D. E. Goldberg, Genetic Algorithms in Search, *Optimization and Machine Learning*, Addison-Wesley Publishing Company, Inc., Reading, MA, 1989.
17. S. Gupta, R. Tiwari, and S. B. Nair, Multi-objective Design Optimization of Rolling Bearings Using Genetic Algorithms, *Mechanism and Machine Theory*, vol. 42, pp. 1418–1443, 2007.
18. T. A. Harris, *Rolling Bearing Analysis*, John Wiley, New York, 2001.
19. F. Hoffmeister and J. Sprave, Problem-independent Handling of Constraints by Use of Metric Penalty Functions, in L. J. Fogel, P. J. Angeline, T. Back (eds.), *Proceedings of the Fifth Annual Conference on Evolutionary Programming*, The MIT Press, San Diego, CA, 1996, pp. 289–294.
20. J. H. Holland, *Adaptation in Natural and Artificial Systems*, University of Michigan Press, Ann Arbor, MI, 1965.
21. F. Herrera, M. Lozano, and J. L. Verdegay, Tackling Real-coded Genetic Algorithms: Operators and Tools for Behavioral Analysis, *Artificial Intelligence Review*, vol. 12, pp. 265–319, 1998.
22. IFORS, *Operational Research Hall of Fame John Von Neumann*, International Transactions in Operation Research, vol. 13, pp. 85–90, 2006.
23. International Standard (ISO:355-1977(E)), *Metric Tapered Roller Bearings: Boundary Dimensions and Series Designation*, Geneva, Switzerland, ISO Publishers, 1977.
24. W. E. Jamison, J. J. Kauzlarich, and E. V. Mochel, Geometric Effects on Rib-roller Contact in Tapered Roller Bearing, ASLE Tranctions, vol. 20, pp. 79–88, 1975.
25. C. L. Karna, Performance Characteristics at the Rib Roller End Contact in Tapered Roller Bearings, ASLE Transactions, vol. 17(1), pp. 14–21, 1973.
26. J. Y. Liu, Analysis of Tapered Roller Bearings Considering High Speed and Combined Loading, *Transactions of ASME, Lubrication Symposium, May 24-26*, 1976.
27. Z. Michalewicz, *Genetic Algorithms + Data Structures = Evolution Programs*, Springer Verlag, New York, 1992.
28. C. A. Moyer and R. E. Mckelvey, A Rating Formula for Tapered Roller Bearing, SAE Transactions, vol. 71, pp. 490–500, 1979.
29. *PSG Design Data Book*, M/s DPV Printers, Coimbatore, India, 2000.
30. S. S. Rao, *Optimisation Theory and Applications*, 3rd ed., John Wiley & Sons, NY, 1996.
31. R. B. Rao and R. Tiwari, Optimum Design of Rolling Element Bearings Using Genetic Algorithms, *Mechanism and Machine Theory*, vol. 42, pp. 233–250, 2007.
32. SKF, *General Catalogue*, Media Print, Paderborn, Germany, 2005.
33. K. Sunil Kumar, R. Tiwari, and R. S. Reddy, Development of an Optimum Design Methodology of Cylindrical Roller Bearing Using Genetic Algorithms, International Journal for Computational Methods in Engineering Science and Mechanics, vol. 9(6), pp. 321–341, 2008.
34. K. Sunil Kumar, R. Tiwari, and P. V. V. N. Prasad, An Optimum Design of Crowned Cylindrical Roller Bearings Using Genetic Algorithms, Transactions of ASME, Journal of Mechanical Design, vol. 131, no. 5, p. 051011, 2009.

APPENDIX A: NOMENCLATURE

A_f	Area of the flange
b_m	A factor in C_d to accommodate modern technological development
B	Width of the cone, mm
$B_{1_{\min}}$	Width of the narrow front face of cone, mm
$B_{2_{\min}}$	Width of the flange, mm
C	Width of the cup, mm
C_d	Dynamic load capacity of the bearing, kN
C_{d_g}	Dynamic load capacity of the bearing obtained using GAs, kN
C_{ds}	Dynamic load capacity of the bearing given in bearing catalogues, kN
$C_{1_{\min}}$	Minimum width of the back-face of the cup, mm
$C_{2_{\min}}$	Minimum width of the front-face of the cup, mm
d	Inner (or bore) diameter of the bearing, mm
d_1	Diameter of the flange-rib of the cone, mm
D	Outer diameter of the bearing, mm
D_m	Bearing pitch diameter, mm
D_r	Roller mean diameter, mm
$D_{r_{L.L}}$	Lower limit of the roller mean diameter, mm
$D_{r_{U.L}}$	Upper limit of the roller mean diameter, mm
$D_{o_{inner}}$	Minimum diameter of the cup, mm
e	Parameter for the mobility condition in Eqs. (27) and (28)
EI	Section modulus of flange section subjected to bending, N-mm^2
$f(X)$	Objective vector
h_f	Position of the rib-roller contact on the flange face, mm
k	Stribeck's constant
$K_{D\min}$	Minimum roller diameter limiter in Eq. (25)
$K_{D\max}$	Maximum roller diameter limiter in Eq. (26)
l	Total length of the roller, mm
l_e	Effective length of the roller, mm

L_{10}	Life of bearing in revolution with 90% reliability
n	Load life exponent
P	Equivalent radial load, N
Q_i	Load on the inner ring at the most heavily loaded roller, N
Q_o	Load on the outer ring at the most heavily loaded roller, N
Q_f	Load on the flange at the most heavily loaded roller, N
r	Corner radius of the roller, mm
r_1	Cone back-face chamfer height, mm
r_2	Cone back-face chamfer width, mm
r_3	Cup back-face chamfer height, mm
r_4	Cup back-face chamfer width, mm
r_5	Chamfer height and width of the front-face the cone and the cup, mm
$S^i_{1_{\min}}$	Minimum thickness of the front-face of the cone, mm
$S^i_{2_{\min}}$	Minimum thickness of the back-face of the cone, mm
$S^o_{1_{\min}}$	Minimum thickness of the back-face of the cup, mm
$S^o_{2_{\min}}$	Minimum thickness of the front-face of the cup, mm
T	Total width of the bearing, mm
X	Design variable vector
Z	Number of rollers
α	Contact angle of the outer raceway (i.e., cup)
α_f	Flange angle
α_i	Contact angle of the inner raceway (i.e., cone)
β	Parameter for the effective length of the roller in Eq. (38)
β^o	Semi-taper angle of the roller
ε	Parameter for the outer ring strength consideration in Eq. (29)
γ	Ratio, $D_r \cos\alpha / D_m$
λ	Reduction factor for the manufacturing and mounting deviations
λ_l	Life factor
ν	Factor to account for edge loading
σ_{bf}	Bending stress in the flange, N/mm^2
$\sigma_{f_{\max}}$	Maximum stress in the flange, N/mm^2
$\sigma^l_{\max}$	Maximum contact stress, N/mm^2
σ_{safe}	Safe contact stress, N/mm^2
σ_{tf}	Direct tensile stress in the flange, N/mm^2
τ_f	Shear stress in the flange, N/mm^2
$\sum\rho$	Curvature sum of contacting bodies
$F(\rho)$	Curvature function of contacting bodies

Subscripts

i	Represents the inner raceway or cone
o	Represents the outer raceway or cup

APPENDIX B: GEOMETRIC RELATIONS OF TAPERED ROLLER BEARINGS

The design of tapered roller bearings requires obtaining the internal geometric variables of tapered roller bearings. Internal geometric variables include basic design variables and other internal geometric variables that can be expressed in terms of boundary dimensions, and also the basic design variables using trigonometric relations. It could be shown that the designer has four basic design variables to be optimized and other internal geometric variables are used to formulate constraints for the optimization problem.

Minimum Thickness of the Front-face of the Cup

From Figure B.1 the minimum thickness of the front-face of the cup can be given as

$$S^o_{2_{\min}} = \frac{1}{2}D - \mathrm{HI} = \frac{1}{2}D - \left\{\frac{1}{2}D_m \operatorname{cosec}(\alpha - \beta^o) + \frac{1}{2}l\right\} \sec\beta^o \sin\alpha \quad \text{(B.1)}$$

Minimum Thickness of the Front-face of the Cone

From Figure B.1 the minimum thickness of the front-face of the cone can be given as

$$S^i_{1_{\min}} = \mathrm{FJ} - \frac{1}{2}d = \left\{\frac{1}{2}D_m \operatorname{cosec}(\alpha - \beta^o) - \frac{1}{2}l\right\} \times \sec\beta^o \sin(\alpha - 2\beta^o) - \frac{1}{2}d \quad \text{(B.2)}$$

Minimum Width of the Front-face and the Back of the Cup

From Figure B.1 the minimum width of the front-face of the cup can be given as

$$C_{2_{\min}} = (C + AX) - AI \quad \text{(B.3)}$$

which can also be written as

$$C_{2_{\min}} = \left[C + \frac{1}{2}D_{o_{\text{inner}}} \cot\alpha\right] - \left[\left\{\frac{1}{2}D_m \operatorname{cosec}(\alpha - \beta^o) + \frac{1}{2}l\right\} \sec\beta^o \cos\alpha\right] \quad \text{(B.4)}$$

From Figure B.1 the total width of the cup can be given as

$$C = C_{2_{\min}} + C_{1_{\min}} + \frac{l\cos\alpha}{\cos\beta^o} \quad \text{(B.5)}$$

From equation (B.5) the minimum width of the back-face of the cup $C_{1_{\min}}$ can be expressed as

$$C_{1_{\min}} = C - C_{1_{\min}} - \frac{l\cos\alpha}{\cos\beta^o} \quad \text{(B.6)}$$

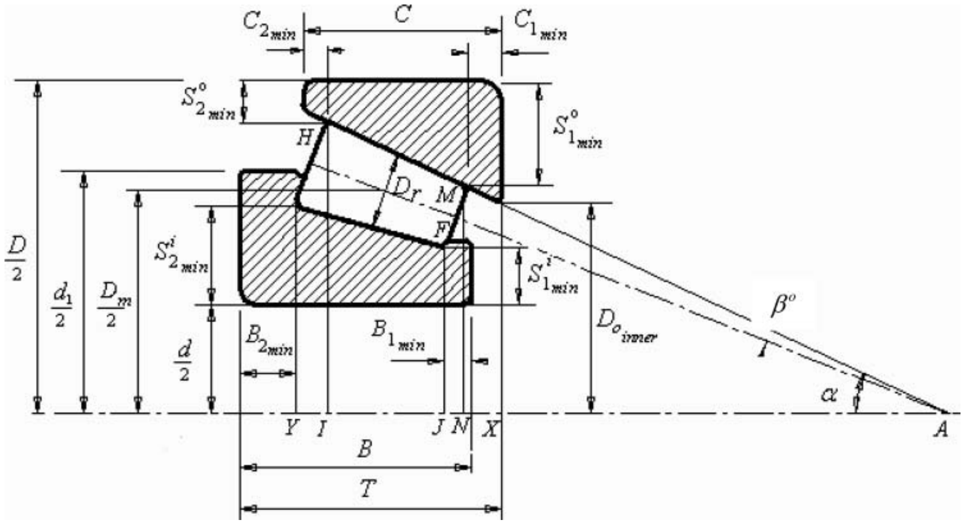

FIG. B.1. The sectional view of a tapered roller bearing with the cup, cone, and roller.

Minimum Width of the Front-face and the Back of the Cone

From Figure B.1 the minimum width of the back-face of the cone can be given as

$$B_{2_{\min}} = (T + AX) - AY \tag{B.7}$$

which can also be expressed as

$$B_{2_{\min}} = \left[T + \frac{1}{2}D_{o_{inner}} \cot\alpha\right] - \left[\left\{\frac{1}{2}D_m \mathrm{cosec}(\alpha - \beta^o) + \frac{1}{2}l\right\} \sec\beta^o \cos(\alpha - 2\beta^o)\right] \tag{B.8}$$

From Figure B.1 the cone width of the bearing (i.e., B) and minimum widths of faces of the cone ($B_{1_{\min}}, B_{2_{\min}}$) can be related with internal dimensions and is given as

$$B = B_{1_{\min}} + B_{2_{\min}} + \frac{l}{\cos\beta^o}\cos(\alpha - 2\beta^o) \tag{B.9}$$

From which the width of the front face of the cone $B_{1_{\min}}$ can be expressed as

$$B_{1_{\min}} = B - B_{2_{\min}} - \frac{l}{\cos\beta^o}\cos(\alpha - 2\beta^o) \tag{B.10}$$

From Figure B.1 the thickness of the back-face of the cup can be given as

$$S^o_{1_{\min}} = \frac{1}{2}D - MN \tag{B.11}$$

which can be written as

$$S^o_{1_{\min}} = \frac{1}{2}D - \left\{\frac{1}{2}D_m - \frac{1}{2}l\sin(\alpha - \beta^o)\right\} + \left\{\left(\frac{1}{2}D_r - \frac{1}{2}l\sin\beta^o\right)\cos(\alpha - \beta^o)\right\} \tag{B.12}$$

Minimum Thickness of the Back-face of the Cone

From Figure B.1 the minimum thickness of the back-face of the cone can be given as

$$S^i_{2_{\min}} = \left\{\frac{1}{2}D_m - \frac{D_r}{2}\cos(\alpha - \beta^o) + \frac{l}{2}\sin\beta^o\right\} - \frac{d}{2} \tag{B.13}$$

APPENDIX C: STRESSES IN FLANGES OF A TAPERED ROLLER BEARING

To calculate stresses in the flanges of a tapered roller bearing, the expressions for reaction loads on the cone, the cup, and the flange are to be considered. The flange of a tapered roller bearing is subjected to the shear, bending, and tensile stresses due to the loads acting on the flange from the roller, and are discussed in this section. The flange force can be calculated from the force equilibrium of the roller. The equilibrium of the roller is shown in Figure C.1.

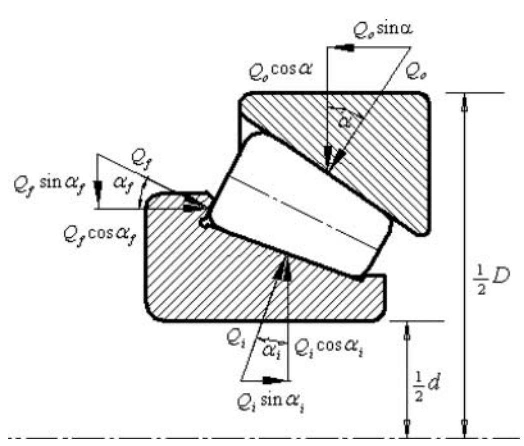

FIG. C.1. The free body diagram of a tapered roller in a tapered roller bearing.

where Q_i is the normal force on the cone, which can be calculated by using the Stribeck's approximate formula [18], Q_o is the normal force on the cup, Q_f is the force on the spherical-face of the roller, α is the nominal contact angle of the bearing, α_i is the contact angle of the cone, and α_f is the flange angle. From the force equilibrium equations we get

$$Q_f = Q_i \cos\alpha_i \frac{(\sin\alpha - \tan\alpha_i \cos\alpha)}{\sin(\alpha + \alpha_f)} \quad \text{and}$$
$$Q_o = Q_i \cos\alpha_i \frac{\left(\sin\alpha_f + \tan\alpha_i \cos\alpha_f\right)}{\sin(\alpha + \alpha_f)} \tag{C.1}$$

Stresses in the Flange

Due to the axial component of the flange force Q_f, the flange is subjected to a direct shear stress τ_f, a bending stress σ_{bf}, and also it is subjected to a direct tensile stress σ_{tf}, due to the radial component of the flange loads. The tensile stress in the flange is given as

$$\sigma_{tf} = \frac{Q_f \sin\alpha_f}{A_f} \quad \text{with} \quad A_f = \frac{\pi\left(\frac{1}{2}d + S^i_{2_{\min}}\right) B_{2_{\min}}}{Z} \tag{C.2}$$

The maximum shear stress in the flange is given as

$$\tau_f = 1.5 \frac{Q_f \cos\alpha_f}{A_f} \tag{C.3}$$

The bending stress in the flange is given as

$$\sigma_{bf} = \frac{Q_f h_f \cos^2\alpha_f}{EI} \quad \text{where} \quad h_f = \frac{\frac{1}{2}d_1 - \left(\frac{1}{2}d + S^i_{2_{\min}}\right)}{2\cos\alpha_f} \tag{C.4}$$

The maximum principal stress in the flange can be obtained using the formula

$$\sigma_{f_{\max}} = \frac{\sigma_{tf} + \sigma_{bf}}{2} + \sqrt{\left(\frac{\sigma_{tf} + \sigma_{bf}}{2}\right)^2 + \tau_f^2} \tag{C.5}$$

On substituting equations (C.2), (C.3), and (C.4) into equation (C.5), the expression for the maximum principle stress in the flange can be obtained and is kept below the yield strength using a constraint, as discussed in Section 3.4.